

Naziru Halilu

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[linkedin.com/in/naziru-halilu-a9a815313](https://www.linkedin.com/in/naziru-halilu-a9a815313) | [GitHub](#) | [Zenodo record list](#)

PERSONAL STATEMENT

Agricultural engineer specialising in UAV and sensor based high throughput phenotyping for cereal and row crops. Builds multispectral and hyperspectral imaging and vegetation index pipelines, self supervised and hybrid deep learning models for crop stress and yield trait detection, and reproducible, uncertainty aware machine learning workflows integrated with GIS, satellite remote sensing, and crop simulation models.

EDUCATION

Erasmus Mundus Joint Master's Degree (EMJMD), Sustainable Agriculture 2025 to 2027

Public University of Navarre, Spain / UTAD, Portugal / Agricultural University of Athens, Greece

BEng (Hons), Agricultural and Bioresources Engineering, First Class 2019 to 2024

Ahmadu Bello University, Zaria, Nigeria. CGPA 4.77/5.00, ranked top 1% of cohort

National Diploma, Agricultural Engineering Technology, Distinction 2016 to 2018

Division of Agricultural Colleges, ABU Zaria. CGPA 3.78/4.00, ranked top 1%

EMPLOYMENT HISTORY

Summer Intern, Computer Vision Group

Jun 2026 to Aug 2026

Dept. of Computer Engineering, ABU, Zaria

- Extended the Geometry Agnostic Contrastive Learning (GACL) framework with four new geometry invariant neural components, reaching 64.4% classification accuracy across 19 disease classes; formalised as a first author paper under review at Biosystems Engineering. [Code](#)

EMJMD Research Scholar (Mobility Track)

Sep 2025 to Jun 2026

Public University of Navarre, Spain and UTAD, Portugal

- Built a 14 module open source pipeline for RGB/multispectral vegetation indexing and CNN/Faster R-CNN inference, reaching 33.8% balanced accuracy. [Code](#)
- Designed STADOS, a five stage radiometric correction framework; temporal NDVI variance fell by up to 39%. [Code](#)
- Built a two layer GIS and regression kriging pipeline near Tudela, Spain; RMSE 7.78, R sq 0.77. [Code](#)
- Integrated GIS, remote sensing, and machine learning across 65+ hectares of Portuguese trial sites, cutting lime demand by 2.69 tonnes. [Code](#)
- Developed Multitask Singularity Regression Kriging (MSRK), cutting residual mapping uncertainty by roughly 20%. [Code](#)
- Built a Bayesian grounded digital twin predicting a 47.8% reduction in annual soil loss for gully erosion control. [Code](#)

Undergraduate Project Assistantships

Mar 2024 to Mar 2025

Depts. of Agricultural and Bioresources Engineering and Animal Science, ABU

- Contributed data collection, statistical analysis, and literature review across concurrent undergraduate research projects spanning water and irrigation systems, soil and erosion assessment, and postharvest and animal systems.

Project Assistant (Remote Data Lead), Smart Greenhouse Automation

2024 to 2025

IAR-ABU, Nigeria

- Automated greenhouse environmental control by engineering adaptive irrigation scheduling algorithms fusing real time soil moisture and microclimate telemetry.

Student Research Assistant, Precision Crop Systems

2022 to 2024

NAERLS, Nigeria

- Trained and validated CNN based crop stress detection models reaching 78% classification accuracy on multispectral UAV imagery; automated UAV flight path planning.

Assistant Farm Manager (concurrent, part time)

2022 to 2024

Yankari Nasarawa Farms, Nigeria

- Sustained animal welfare and operational efficiency across a commercial poultry facility by managing automated climate control and water dosing systems.

Project Assistant, Smart Irrigation and Water Systems

2021 to 2022

BOMO Basin Project, Nigeria

- Automated water delivery across a one hectare smart drip irrigation array by implementing closed loop control logic.

Research Intern

2017 to 2020

NCAM, Ilorin and IAR, Zaria, Nigeria

- Benchmarked mechanisation efficiency of locally engineered machinery via Python data logging pipelines; supported baseline hydrological research.

PUBLICATIONS

Peer Reviewed

- Halilu, N., Sulaiman, J. (2026). Real Time RGB Proxy Vegetation Indexing and Texture Analysis for UAV and Handheld Crop Imagery. IJRSI, 13(7), 2951 to 2989. [DOI](#) | [Code](#)
- Halilu, N., Sulaiman, J. (2026). Sensing the Invisible: A Critical Review of NIR and Hyperspectral Analysis for Crop and Food Quality Assessment. IJRSI, 13(7), 2990 to 3020. [DOI](#)
- Ismail, K., Halilu, N. (2024). AI Driven Precision Agriculture System for Optimizing Maize Yield in Northern Nigeria. Proc. AfCPA, 2(1), 100 to 120.
- Halilu, N. (2026). A Hybrid CNN-BiLSTM-Attention Deep Learning Framework for Sediment Transport in an Agricultural Gully System. IJRSI, 13(7), 2920 to 2950. [DOI](#) | [Code](#)
- Naziru, H., Auwal, M., Yusuf, A. (2020). Physio chemical and bacteriological quality of water for irrigation purpose. Nigerian Research Journal of Engineering and Environmental Sciences, 5(1), 198 to 204.

Submitted and Under Review

- Halilu, N., Sulaiman, J. Geometry Agnostic Contrastive Learning (GACL). Biosystems Engineering, Elsevier. Under review. [DOI](#) | [Code](#)
- Halilu, N., Sulaiman, J. Operationalizing GIS and Machine Learning across Contrasting Cropping Systems. European Journal of Agronomy, Elsevier. Under review. [DOI](#) | [Code](#)
- Halilu, N. (2026). Digital Twin for the Evaluation of Experimental Gully Biocontrol Using Morning Glory (Ipomoea spp.). Environmental Monitoring and Assessment, Springer Nature. Under review. [DOI](#) | [Code](#)

Preprints

- Halilu, N. (2026). Multitask Singularity Regression Kriging for Spatio Temporal Soil NPK Crop Stress Yield Prediction. Zenodo Preprint. [DOI](#) | [Code](#)
- Halilu, N. (2026). STADOS: Spectral Temporal Adaptive Dark Object Subtraction. Zenodo. [DOI](#) | [Code](#)
- Halilu, N. (2025). A Systematic Review of IoT, Edge Computing, Time Series Modelling and Control Systems in Precision Irrigation. Zenodo. [DOI](#)

RESEARCH SOFTWARE AND OPEN DATASETS

- Geospatial Data Analysis: six notebook vector/raster GIS, Sentinel 1/2, and Google Earth Engine curriculum. [DOI](#) | [Code](#)
- Integrated Decision Support Model for Dairy Processing: Ridge regression optimiser, R sq 0.859 (LOOCV). [DOI](#) | [Code](#)
- GIS Based Delineation of Areas Suitable for Livestock Slurry Application: RMSE 7.78, R sq 0.77 across 14.35 km sq. [DOI](#) | [Code](#)
- NaCROP (Zaria Crop ET and Irrigation DSS): desktop irrigation scheduling tool, water balance closure within 0.1 mm. [Code](#)

LEADERSHIP AND VOLUNTEERING

Undergraduate Course Tutor

2020 to 2024

ABU Zaria (NAMET), Nigeria

- Ran supplementary tutorials in Differential Equations and Transforms, Irrigation System Design, and Computer Applications in Agricultural Engineering.

Secretary General, NAAESB

2023 to 2024

National Assoc. of Agric. Engineering Students

- Directed operational targets and inter chapter scientific meetups for the national student body.

Deputy Coordinator

2020 to 2025

KOJYAF, Zango Shanu, Zaria

- Coordinated tutorial and mentorship programmes preparing secondary students for JAMB entrance examinations.

Director, Environmental Protection Agency

2023 to 2024

ABU Students' Union Government

- Directed campus environmental protection and sanitation initiatives.

Physics and Mathematics Teacher

2022 to 2023

Takai Academy, Zaria, Nigeria

- Instructed senior classes in classical physics and computational principles.

KEY SKILLS

Geospatial AI / GIS

QGIS, ArcGIS, ENVI, Google Earth Engine, Sentinel imagery, UAV imaging

Machine Learning	PyTorch, TensorFlow, Keras, scikit-learn, CNNs, self supervised learning
Crop Modelling	APSIM Next Generation, crop growth simulation
Climate AI	CMIP6, ERA5, ACCESS-S2, downscaling and forecasting
3D Computer Vision	COLMAP, Open3D, Pix4D, Agisoft Metashape
Programming	Python, MATLAB, R, Git/GitHub, Zenodo

CONFERENCES AND WORKSHOPS

Digital Agroecology Cluster Conference (EU funded), Brussels (2025); Vibrational Spectroscopy and Chemometrics in Agro Food Systems (EU funded) (2025); IEEE AgriTech, Valencia, Spain (2024); 6th African Regional Conference on Irrigation and Drainage (2025); Mechanistic Modeling of Rainfall Driven Soil Solute Transport, University of Florida (2025); NAERLS Technical Seminar Series (2023).

AWARDS AND HONOURS

Erasmus Mundus Joint Master Scholarship, European Union (2025); Best Project Award, ABU Senate (2024); Best Graduating Student, Agricultural and Bioresources Engineering (2024); Nigerian Federal Government Award for Academic Excellence (2024); Best Undergraduate Research Paper Award, ABU (2024); Best 300 Level Student, ABU (2022); Best 200 Level Student, ABU (2021); Award of Merit as Tutor, Faculty of Engineering, ABU (2021); Best Graduating Student, National Diploma, Samaru College of Agriculture (2019); PTDF Undergraduate Fellow (2023); Needs Based University Fellowship (2022).

PROFESSIONAL MEMBERSHIPS

American Society of Agricultural and Biological Engineers (ASABE); Nigerian Institution of Agricultural Engineers (NIAE); Nigerian Society of Engineers (NSE); Pupil Engineer, COREN.

REFERENCES

Prof. Miguel Ángel Campo Bescós, Public University of Navarre, miquel.campo@unavarra.es, +34 948 16 9077
 Prof. José Tadeu Marques Aranha, UTAD, Portugal, j_aranha@utad.pt, ext. 4493
 Dr. Ibrahim Babangida Dalha, Ahmadu Bello University, ibdalha@abu.edu.ng, +234 806 687 3858